

Design and development of the control PCB for a drone with a study of potential space applications

Master's degree in Aerospace Science and Technology (MAST)

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1 - Objectives and schedule

Weeks			Objective	State	Description/Others
1	Class	2025-09-08	Make introduction	Achieved	Introduction done
2	Class	2025-09-15	Expose some examples	Half-achieved	Exposed some examples, but not all them.
3	Class	2025-09-22	Finish Intro-Start selecting the components	Half-achieved	Into finished, some components are selected + altium course almost done
4	Class	2025-09-29	Select the components	Half-achieved	Altium education course done + manufacturer selected + main components selected
5	Class+	2025-10-06	Make a PCB (schematic)	Not achieved	ESA week, not too much development.
6	Class	2025-10-13	Make a PCB (schematic + routing)	Half-achieved	Component import done but at mid development of schamtics
7	Exams	2025-10-20	Make a PCB (schematics + routing)	Half-achieved	All schematics done
8	Exams	2025-10-27	Make a PCB (routing + manufacturer)	Half-achieved	PCB done
9	Class	2025-11-03	Check Ardupilot/Space applications + Manufacturer	Not achieved	Still developing the PCB because of flaws.
10	Class	2025-11-10	Check Ardupilot/Space applications	Achieved	PCB sent to manufacturer + Checking Space applications
11	Class	2025-11-17	Check Ardupilot/Space applications	Achieved	Checking space requeriments and applications (standards)
12	Class	2025-11-24	Check Ardupilot/Space applications	Half-achieved	Ardupilot operating system operational (still developing bootloader)
13	Class+	2025-12-01	Test PCB (estimated day PCB received)	Not achieved	CNES week, no too much development. PCB received at last moment, preparing equipment.
14	Class	2025-12-08	Test PCB	Achieved	All equipment prepared. PCB electrically stable and STM32 reponse. Checking for ICs reponses and configuration.
15	Class	2025-12-15	Test PCB + Space applications	Achieved	Development of Space electronics and space applications.
16	Freedom	2025-12-22	Test PCB + Space applications	Achieved	Development of Space electronics and space applications.
17	Freedom	2025-12-29	Test PCB + Space applications	Achieved	Space electronics and space applications done. Development of software.
18	Exams	2026-01-05	Test PCB	Achieved	Clock changed (previous clock selection was incorrect), bootloader created for 24mhz. PCB routing chapter finished.
19	Exams	2026-01-12	Test PCB	Achieved	Firmware+Bootloader created and working. 3D structure. Almost all basic test have been performed.
20	Exams	2026-01-19	Test PCB	Achieved	Mounted on a drone. HSE changed, it was broken by an ESD. Configuring for flight. 3 flights performed
21	Freedom	2026-01-26	Test PCB + do last modifications	Achieved	Finishing TFM.
22	Freedom	2026-02-02	Deposit TFM	Achieved	-
23	Freedom	2026-02-09	Expose	Achieved	-

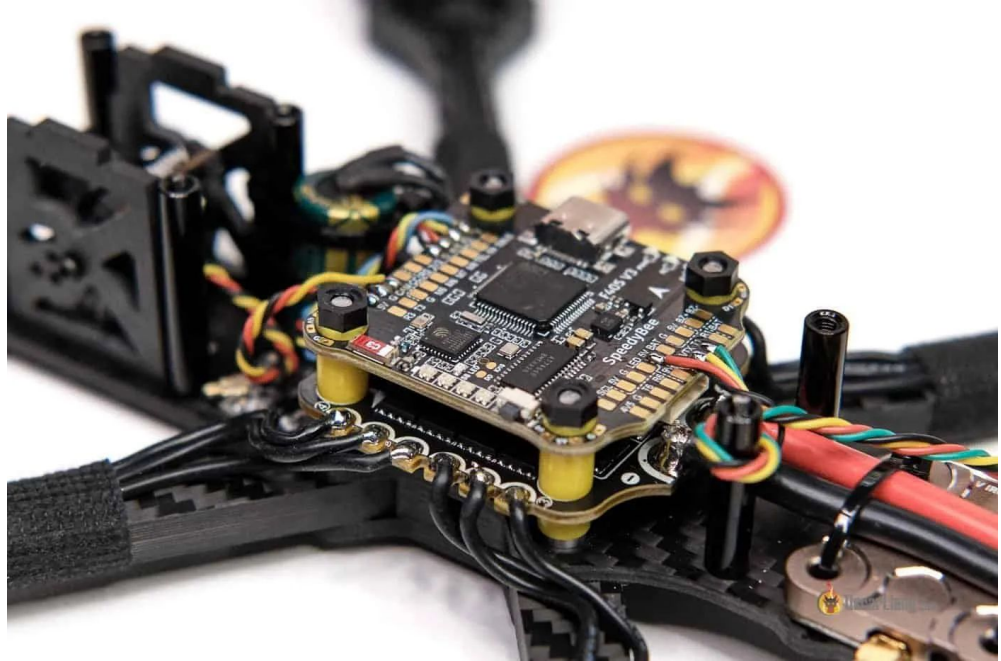
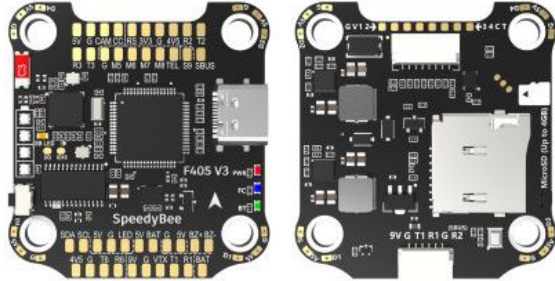
1 - Objectives and schedule

- **Objectives of the TFM:**

- Develop a functional drone control PCB.
- Create the drivers to manage the drone control PCB with ArduPilot.
- Evaluate the current design and expose space requirements.

1 - Objectives and schedule

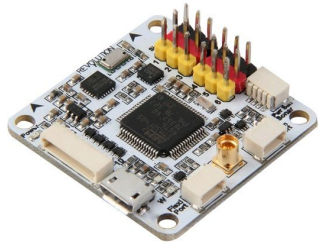
- **Example of commercial drone control PCB.**
 - o Speedybee FC 30-60€



1 - Objectives and schedule

- Other examples:

- OpenPilot Revolution 60€
- Matek FC H743 70-110€
- Pixhawk 60-100€
- CubePilot Cube Orange/+ 350-600€
- CUAU X7 PRO+ 860€
- Makerfire UTMSYS USX51 600-1000€



OpenPilot



Matek



Pixhawk



CubePilot



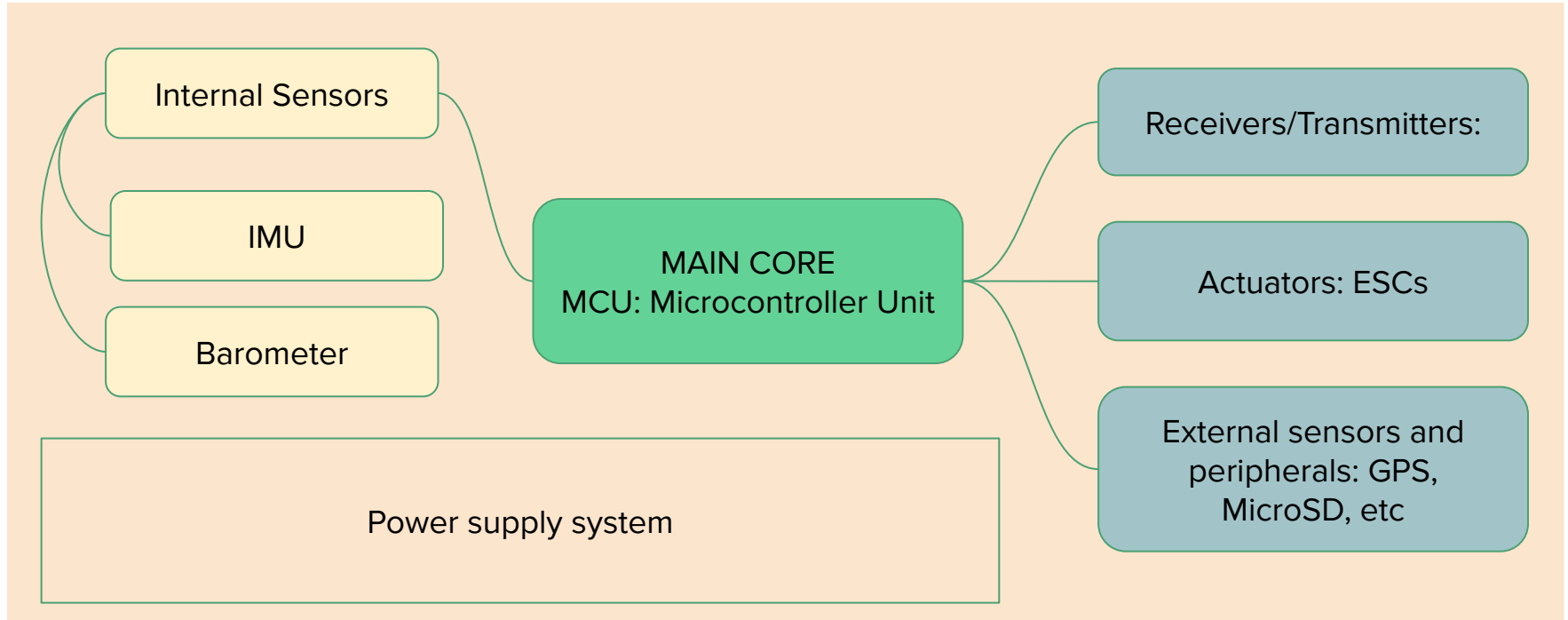
Makerfire



CUAV X7

2- Introduction

- **Basic components of a FC:**



2 - PCB Schematics

- **Before component selection:**
 - Drone Software: Ardupilot (professional firmware) → relies on ChibiOS.
 - ChibiOS is a RTOS (Real Time Operating System).
 - **IMPORTANT:** Not all sensors and MCUs are supported by default.
 - Drone Standard: Pixhawk Standard
 - Pixhawk connector-bus standard applied



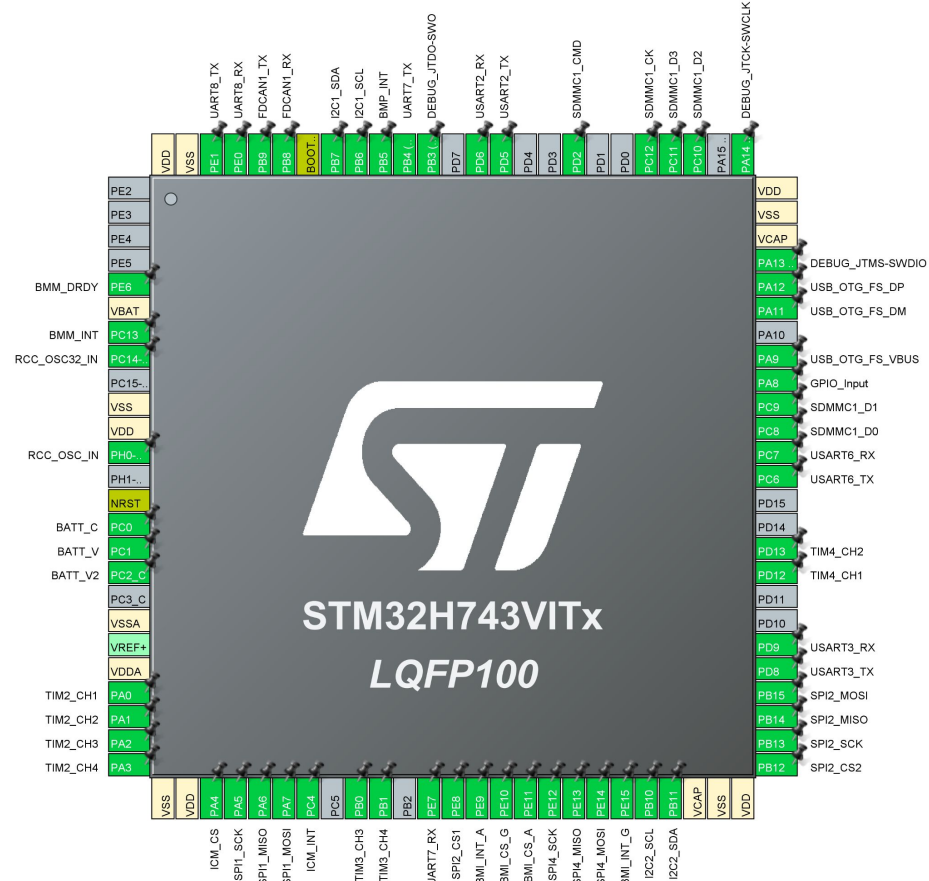
ChibiOS/RT
A compact and fast
real-time operating
system

2 - PCB Schematics

- MCU:

- High performance MCU: STM32H743VIT6

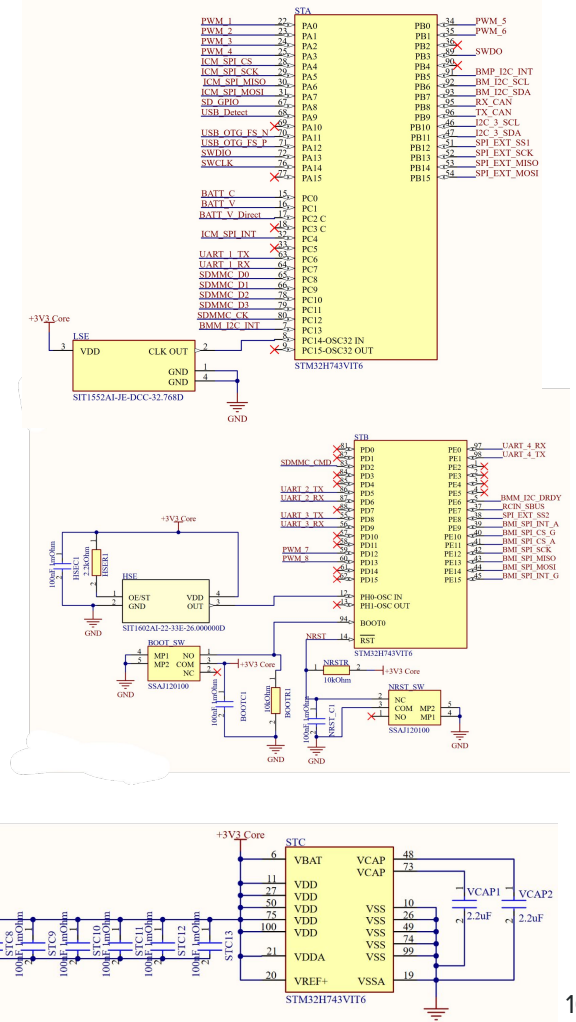
- ARM-M7 RISC 480 MHz
- 2 MB Flash, 1 MB RAM
- LQFP100 (80/100 pins used)
 - 3 SPI (2 internal)
 - 1 SDMMC
 - 2 I2C (1 internal)
 - 3 ADCs (1 internal)
 - 8 PWM
 - 1 CAN
 - 1 USB
 - 1 SWD
 - 4 UART
 - MCU power connections



2 - PCB Schematics

- MCU clock and schematics:

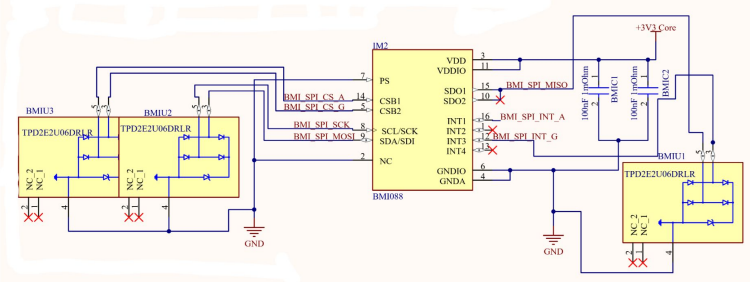
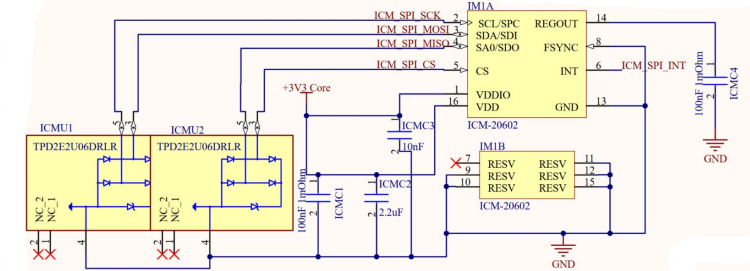
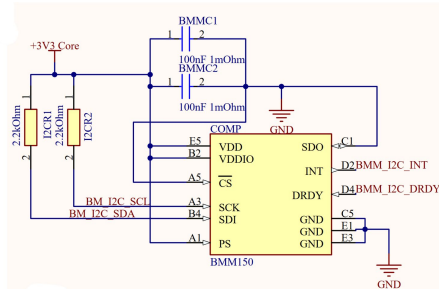
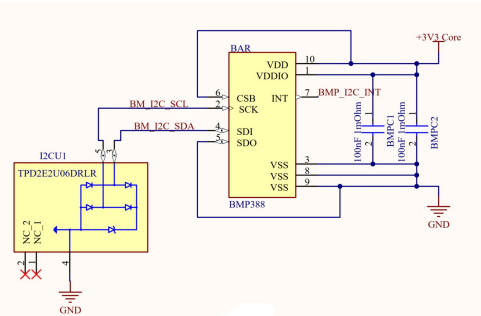
- HSE: ASEMB-24.000MHZ-LC-T (24 MHz)
 - MEMS (Micro ElectroMechanical System) 50 ppm.
 - Good against vibrations.
 - Previously SIT1602-22-33E-26.000000D (26 MHz).
- LSE: SIT1552AI-JE-DCC-32.768D (32.768 KHz)
 - MEMS+TCXO (Temperature Compensated Crystal Oscillator) 5 ppm.
 - Smallest programmable clock in the world.
 - Good against vibrations and temperature variations.
- Conditioning specified as in the datasheet for all components. 1 switch for BOOT and 1 switch to reboot.



2 - PCB Schematics

- Sensors:

- Powered at 3.3 V with ESD-TVES protection with low capacitance 1.5 pF for fast data transmission.
- IMUs (isolated SPIs at 2-8 MHz):
 - ICM-20602 (accel + gyro)
 - BMI088 (accel + gyro)
- Barometer and magnetometer:
 - Shared I2C at 400 kHz, 2.2k Ω pull ups.
 - BMP388 and BMM150



2 - PCB Schematics

- Complex buses/connectors:

- CAN:

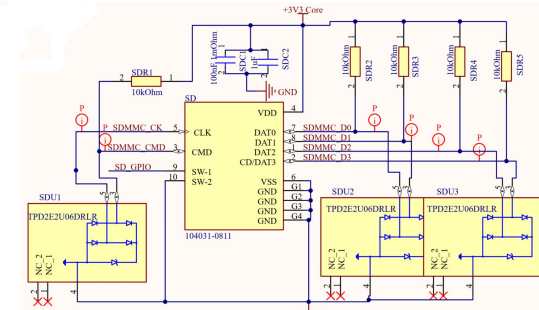
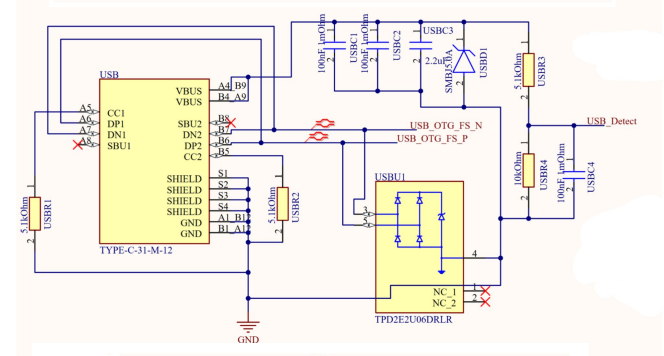
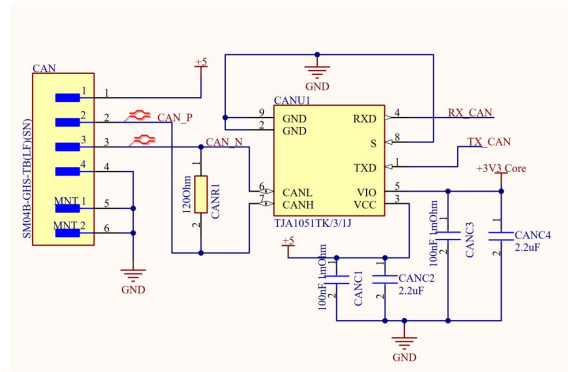
- 120 Ω
- Pixhawk connector SM04B-GHS JST.
- TJA1051TK/3/IJ transceiver (allows 3V3).

- USB 2.0 FS:

- 90 Ω
- 5.1 k Ω for termination device.
- ESD protection + filtering + detection line.
- It does not provide power source.

- SDMMC (MicroSD):

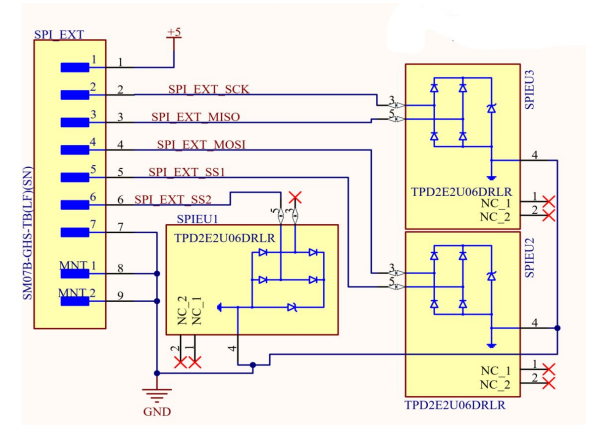
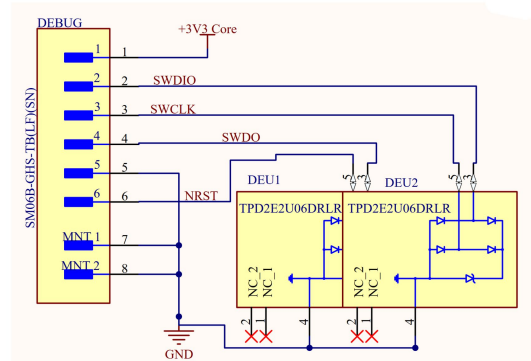
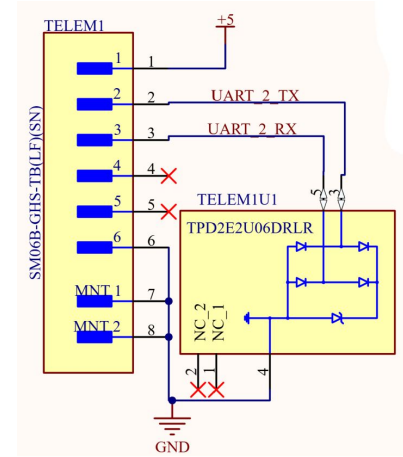
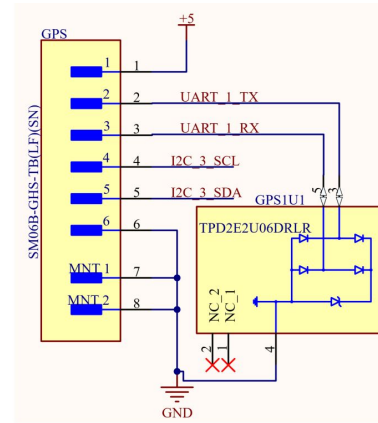
- 10 k Ω pull ups, except on CK line
- ESD protection + filtering + detection line.



2 - PCB Schematics

- Other buses and connectors:

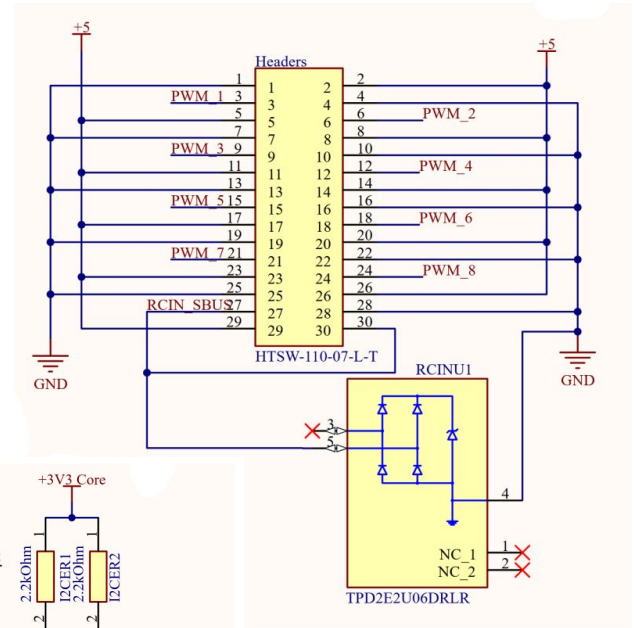
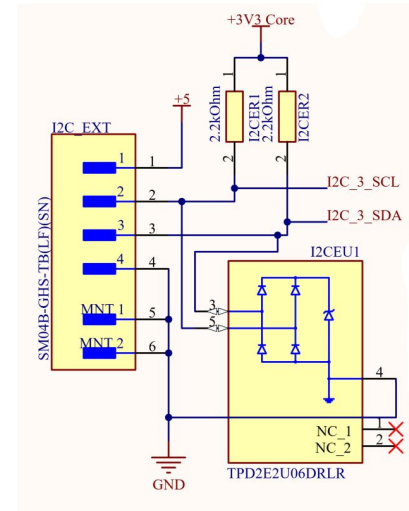
- 1 SWD (debug port): custom connector.
- 1 SPI: pixhawk standard.
- 2 Telem: (UART) reduced version of pixhawk standard.
- 2 GPS: (UART + I2C) reduced version of pixhawk standard.



2 - PCB Schematics

- Other buses and connectors:

- 1 I2C: pixhawk standard.
- 2.54 mm pitch headers/pins:
 - 8 PWM (No ESD protection and no resistance in serie)
 - 1 RCIN conector (UART RX) (admits SBUS with internal inversion).

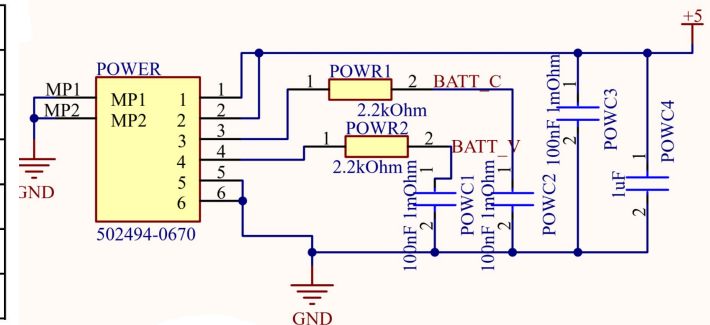


2 - PCB Schematics

- Power supply system:

- Second option: Pixhawk battery connexion port
 - 5.3 V at 2.5 A (real values are near 5.4 V).
 - Filtered.
 - Capable of measure current and voltage from external block.

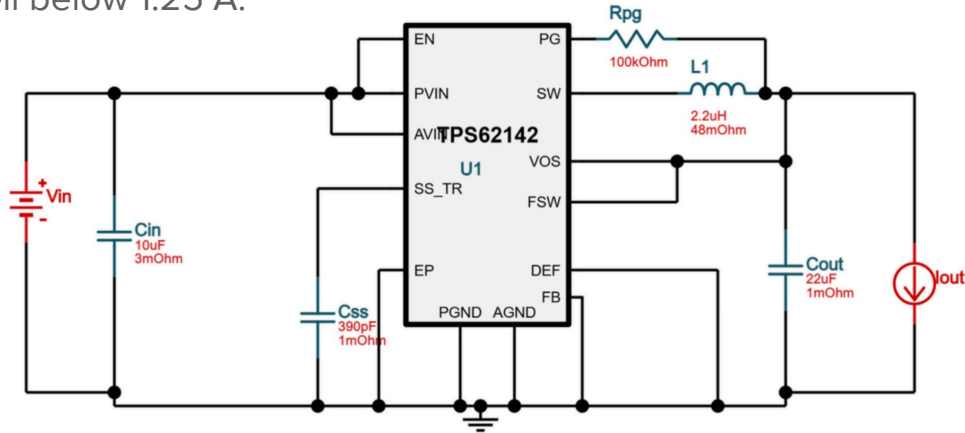
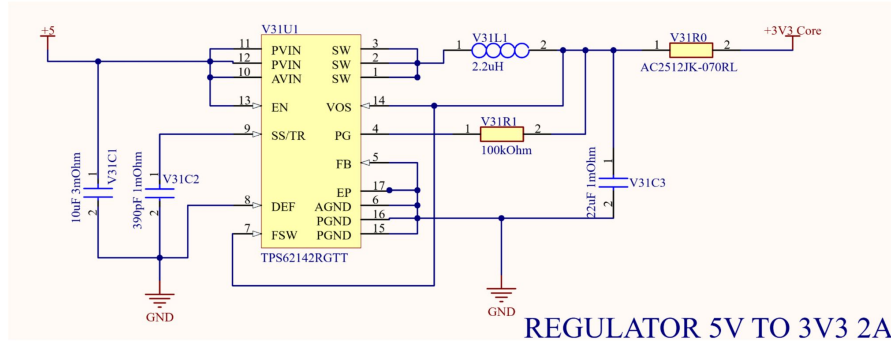
Pin	Signal	Voltage
1 (red)	VCC	+5.3 V
2 (blk)	VCC	+5.3 V
3 (blk)	Current (analog)	+3.3 V
4 (blk)	Voltage (analog)	+3.3 V
5 (blk)	GND	GND
6 (blk)	GND	GND



2 - PCB Schematics

- Power supply system:

- 3.3 V: Dual power regulator for internal and external electronics
 - 2 Isolated power supplies (1 for the board and 1 for peripherals)
 - Using Webench (Texas Instruments), +90 % efficiency.
 - 3.8-6 Vin, 3.3 V 2 A out.
 - Maximum internal current draw well below 1.25 A.



3 - PCB Routing

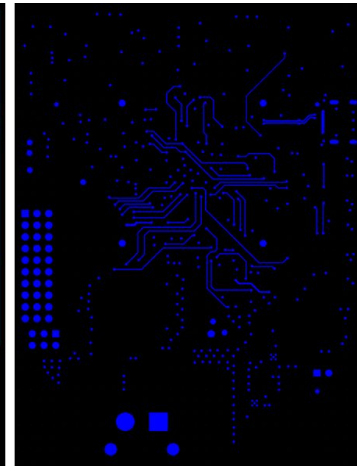
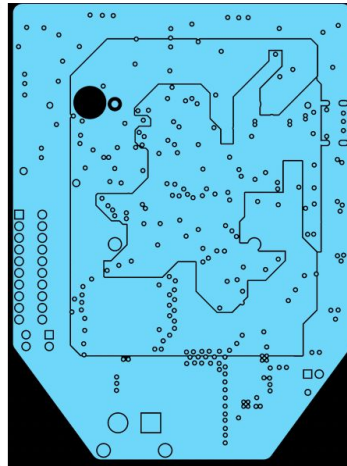
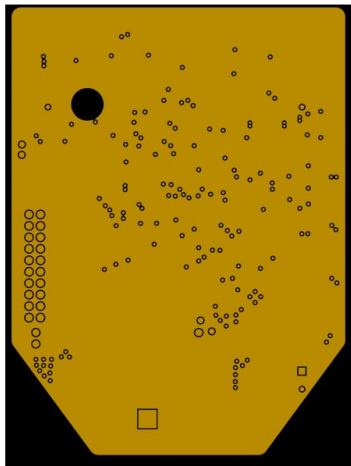
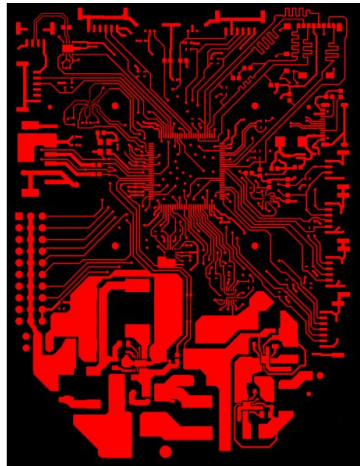
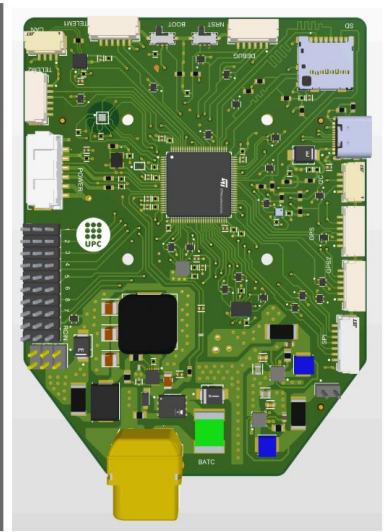
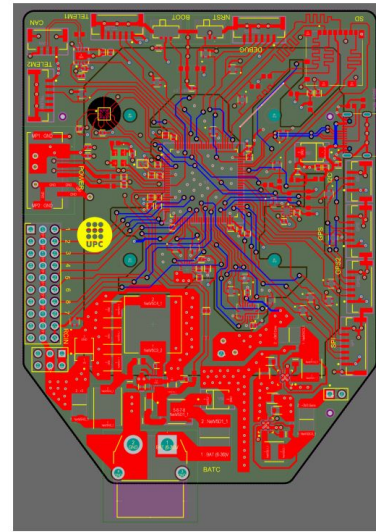
- Layer Stackup JLCPCB:
 - Manufacturer Standard
 - Low cost
 - 4 Layers
- Impedance:
 - 120 Ω CAN bus
 - 90 Ω USB bus

Layer	Material	Thickness (mil)	Thickness (mm)
L1	Outer Copper Weight1oz	1.38	0.0350
Prepreg	7628, RC 49%, 8.6 mil	8.28	0.2104
L2	Inner Copper Weight	0.60	0.0152
Core	1.1mm H/HOZ with copper	41.93	1.0650
L3	Inner Copper Weight	0.60	0.0152
Prepreg	7628, RC 49%, 8.6 mil	8.28	0.2104
L4	Outer Copper Weight1oz	1.38	0.0350

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width	Trace Spacing
120	Differential Pair (Non coplanar)	L1	/	L2	7.1700	14.0000
90	Differential Pair (Non coplanar)	L1	/	L2	11.2600	8.0000
90	Differential Pair (Non coplanar)	L4	L3	/	11.2600	8.0000

3 - PCB Routing

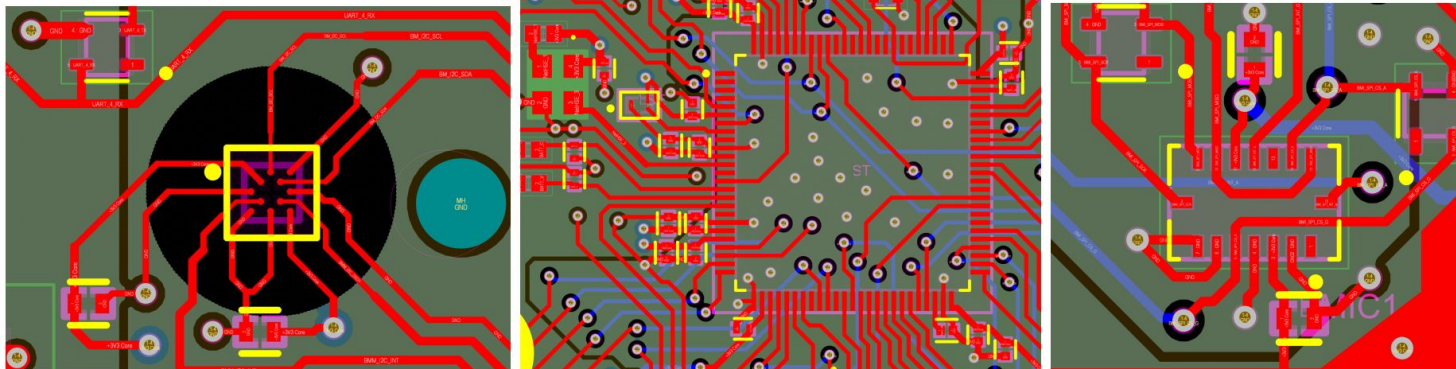
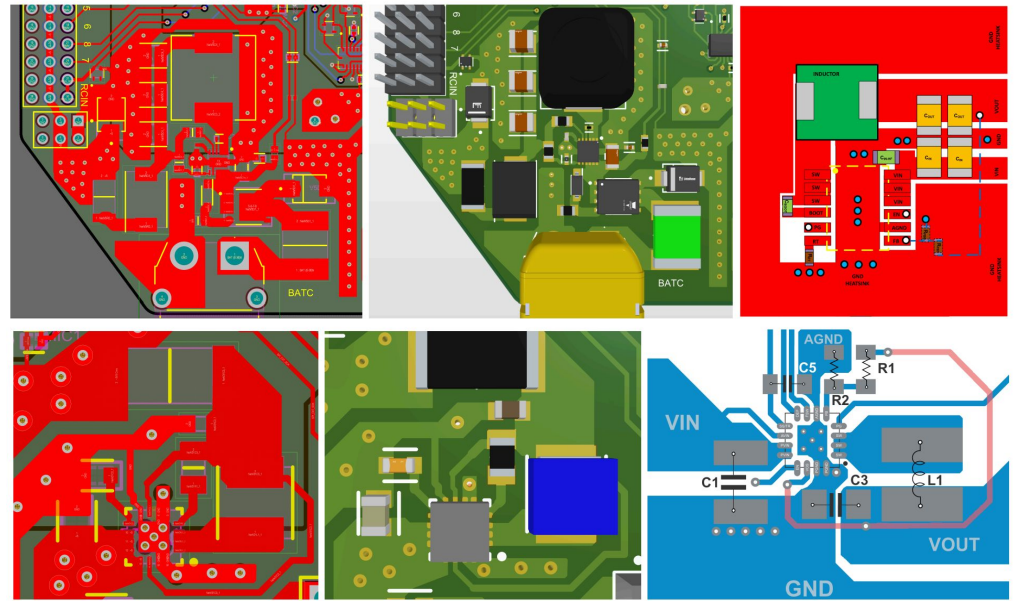
- Planes and trace width IPC-2152:
 - L1: Signal and power lines
 - L2: GND → reference plane
 - L3: Power (5V and 3V3) and GND → reference
 - L4: Signal



Current (A)	Trace Width (mil)
1	10
2	30
3	50
4	80
5	110
6	150
7	180
8	220
9	260
10	300

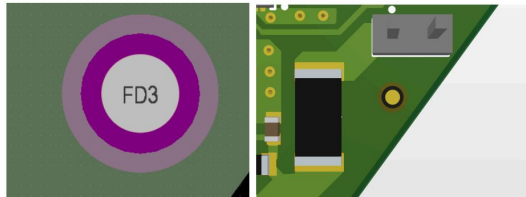
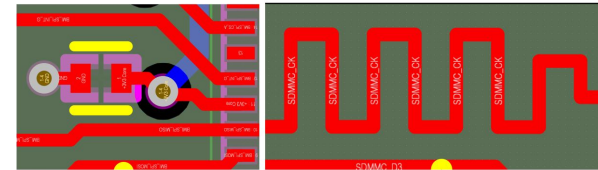
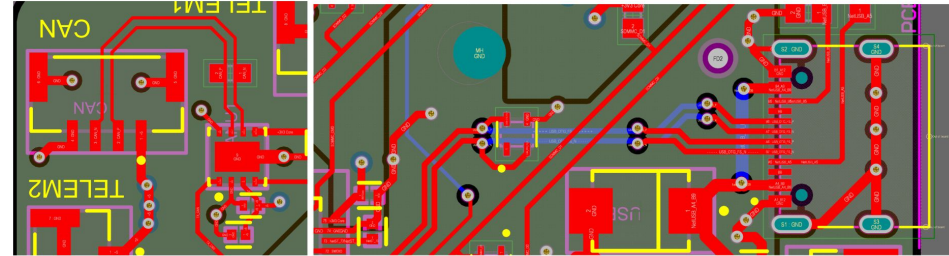
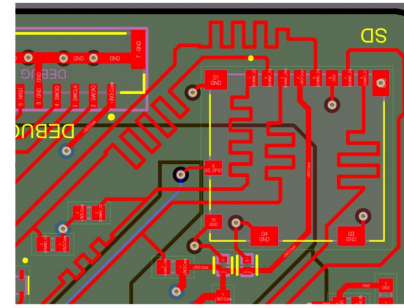
3 - PCB Routing

- Power lines:
 - Following datasheet routing.
 - Sensors and MCU:
 - BMM150 copper pour.
 - Capacitors near MCU.
 - BMI088, merge MISO lines.
- Problematic on high-freq.



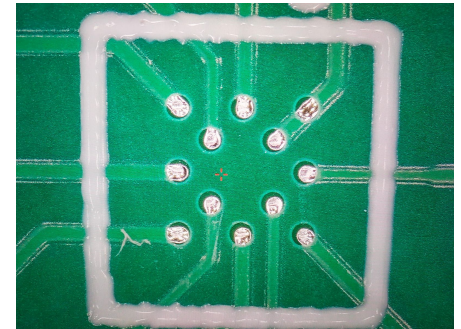
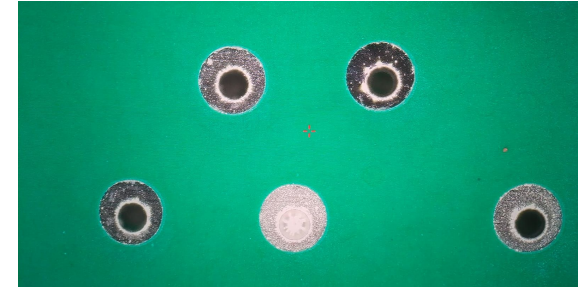
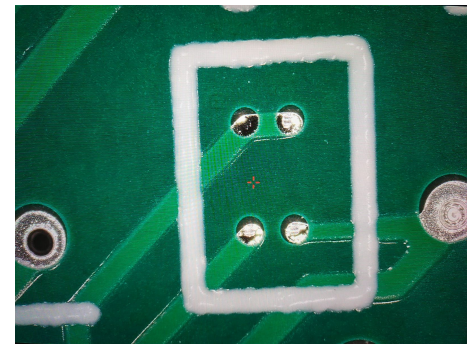
3 - PCB Routing

- Special buses:
 - SDMMC (MicroSD): high-frequency and length matching.
 - USB bus: use of vias to prioritize SDMMC.
 - CAN bus
- Fabrication markers and errors
 - Fiducial
 - Corners, crosstalk and bad capacitor positioning.



4 - Manufacturing and PCBA Files

- Manufacturer most stringent capability and verification:
 - Trace width 4 mil $\pm 20\%$ (39.37 mil = 1 mm) (specially for magnetometer)
 - Un-plug vias for testing and 0402 min size to reduce cost.
 - Always checking DFM (Design For Manufacturability).
 - Altium design rule checker and/or Manufacturer programme.
 - Actual JLCPCB trace positioning deviation < 1 mil. No solderwicking, nor soldebridging.
 - Some vias plugged with solder due to HASL bath.
- Files:
 - Gerber X2
 - CPL (or PnP): Customized (wrong rotations).
 - 146 components per board.
 - BOM: Customized (Manufacturer part number + PCB manufacturer number)
 - 44 different components per board (in fact 50 considering non-used).



5 - Software Porting

- Environment installation:
 - Linux and Ubuntu
 - Download ArduPilot
 - Install ArduPilot and ChibiOS
 - Update them
- Create a folder and add hwdef.dat and hwdef-bl.dat files.
- Create the firmware and the bootloader.

```
wsl --install
wsl --install -d Ubuntu
sudo apt update && sudo apt install git -y

git clone https://github.com/ArduPilot/ardupilot.git

cd ardupilot
git submodule update --init --recursive
Tools/environment_install/install-prereqs-ubuntu.sh -y
. ~/.profile
```

```
cd ardupilot/libraries/AP_HAL_ChibiOS/hwdef
```

```
cd ~/ardupilotcd
ardupilot/libraries/AP_HAL_ChibiOS/hwdeft
./Tools/scripts/build_bootloaders.py CustomH7
./waf configure --board CustomH7
./waf copter
```

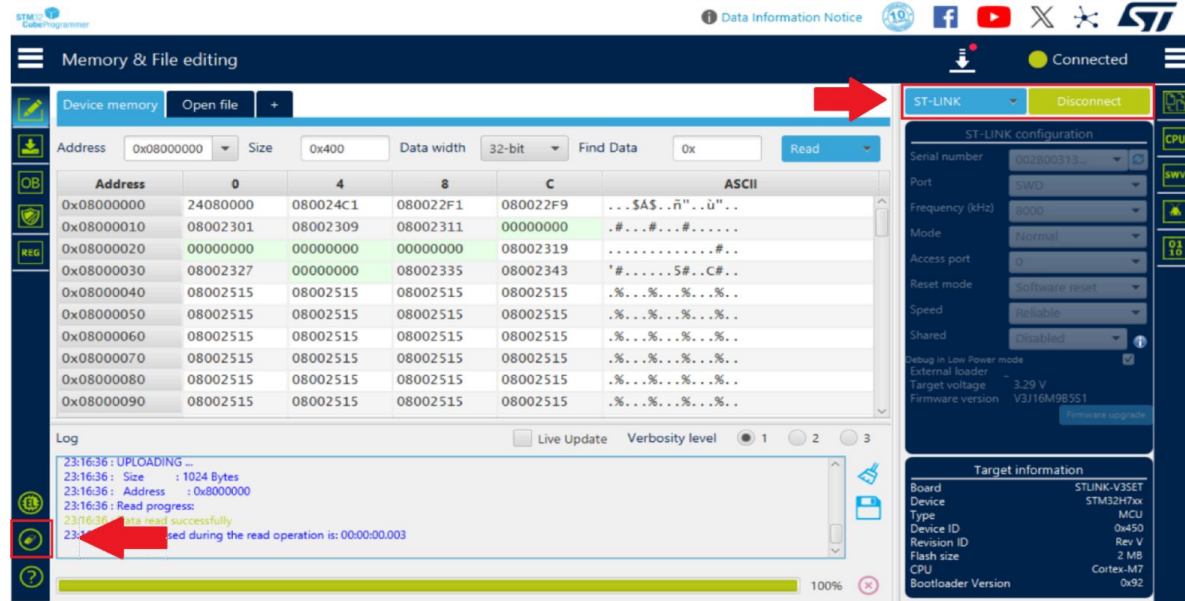
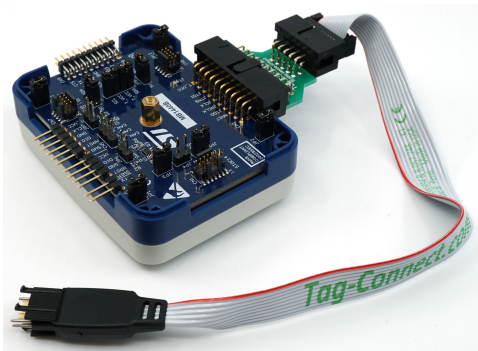
5 - Software Porting

- 2 MB in order:
 - Divided in 16 pages of 128 KB
 - Bootloader 128 KB
 - Dedicated storage 32 KB
 - Firmware + addons 1888 KB
- hwdef.dat and hwdef-bl.dat files:
 - Test-fail-read_doc-try_again
- PLL with 26 MHz HSE:
 - In STM32cubeIDE impossible with auto-mode, possible with manual mode with multipliers/dividers.
 - ChibiOS needs 24 MHz , (Impossible to interact with manual PLL values at 26 Mhz, I tried, plausible reason→USB 48 Mhz), (replacement!!).

```
# Core
MCU STM32H7xx STM32H743xx
FLASH_SIZE_KB 2048
FLASH_RESERVE_START_KB 128
...
# Clock
OSCILLATOR_HZ 24000000
STM32_HSE_BYPASS
STM32_LSE_BYPASS
STM32_SYSCLK 400000000
STM32_ST_USE_TIMER 5
...
# Pins and timers to provide high speed data, even Dshot
PB8 CAN1_RX CAN1
PA0 TIM2_CH1 TIM2_PWM(1) GPIO(50)
...
# Order of buses: (0, 1, 2, 3, 4, 5, ..)
SERIAL_ORDER OTG1 USART6 USART2 USART3 UART7 UART8
...
# Sensor bus speed definition range
SPIDEV icm20602 SPI1 DEVID1 ICM_CS MODE3 2*MHZ 8*MHZ
...
#Sensor drivers and rotation
IMU Invensense SPI:icm20602 ROTATION_YAW_90
...
# System PID page storage and size to store values
STORAGE_FLASH_PAGE 14
define HAL_STORAGE_SIZE 32768
...
# Bus DMA priority (in fact 2 DMAs using DMAMUX)
DMA_PRIORITY SPI1* SPI4* CAN1* SDMMC* USART6*
```

5 - Software Porting

- ArduPilot proposes dfu-utils and the process of porting is not very clear, so:
- STM32cubeProgrammer
- STLINK-SET v3 via SWD
- BOOT0 pin on high
- NRST/RESET down
- Delete previous data

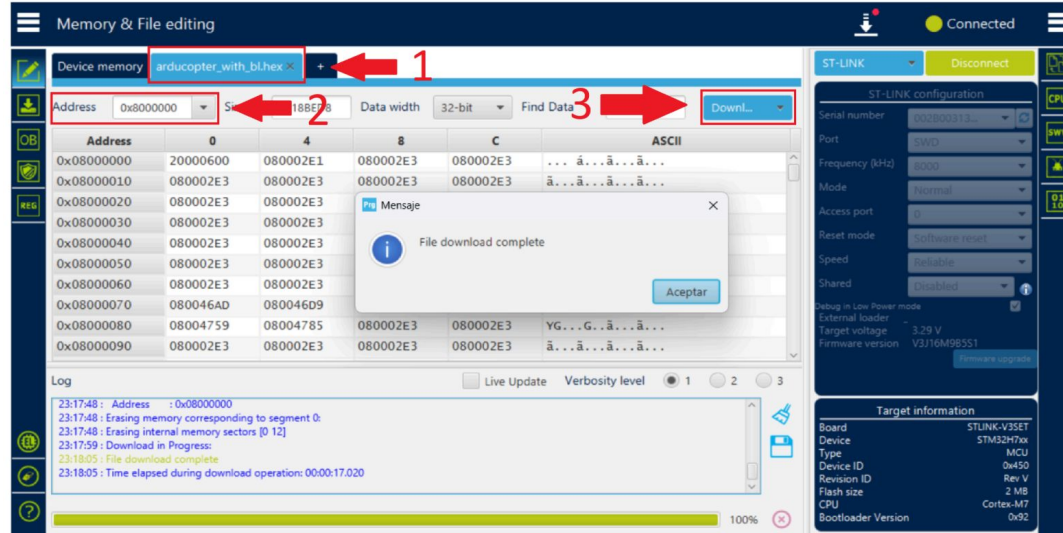


5 - Software Porting

- Delete previous data
- Open the generated code arducopter_with_bl.hex
 - At: ardupilot/build/CustomH7/bin/
- Upload code at: 0x0800 0000

Table 7. Flash memory and SRAM memory mapping for STM32H742xI/G

	Memory	Size in Kbytes	Start address
RAM area	Backup SRAM	4	0x3880 0000
	SRAM4	64	0x3800 0000
	SRAM2	16	0x3002 0000
	SRAM1	32	0x3000 0000
	AXI-SRAM	384	0x2400 0000
	DTCM	128	0x2000 0000
Code area	System memory 2	-	0x1FF4 0000
	System memory	-	0x1FF0 0000
	Flash memory	2048	0x0800 0000
	ITCM	64	0x0000 0000



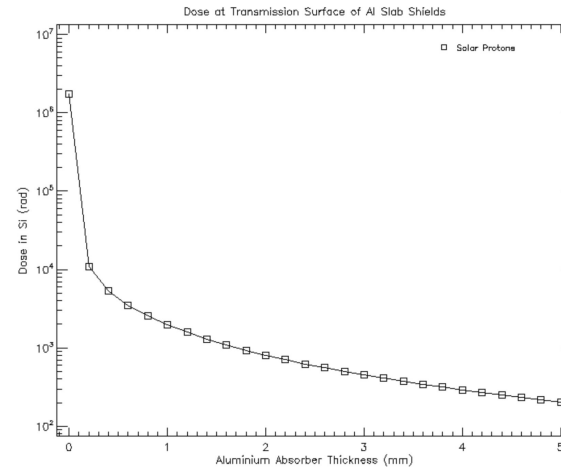
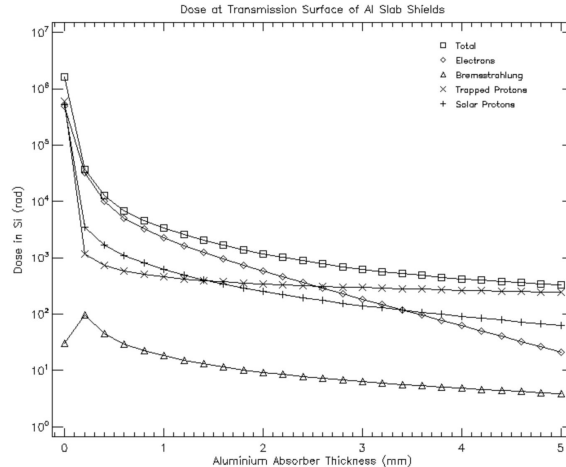
- Lastly reboot the board and BOOT0 pin down.

6 - Space Overview and application

- Possible applications for drones in space.
 - **ISS (inside):** monitor data, evaluate the environment, and assist with daily tasks.
 - Problem → Microgravity (temperature).
 - **MARS:** transport of goods, mapping, data relay, differential positioning, etc.
 - Problem → Low air density, and extreme temperature.
 - **VENUS:** similar applications compared to MARS.
 - Problem → Extreme pressure, temperature and corrosion. Feasible at high altitudes.
 - **TITAN:** similar applications compared to MARS.
 - Problem → Quite feasible, though low temperature.
 - **Other mandatory considerations:** radiation, outgassing, vibrations, shocks, etc.

6 - Space Overview and application (SPENVIS)

- Simple radiation analysis of **STM32H743VIT**, COTS, **2 year mission**:
 - With ECSS **RDM** (Radiation Design Margin) of **2**, and TID of **47 krad** from STM32H753
 - **SSO** (Sun-Synchronous Orbit) 700 km → **0.5 mm** approx minimum of Al protection.
 - **MARS** at 100 km from surface → **0.02 mm** approx minimum of Al protection.

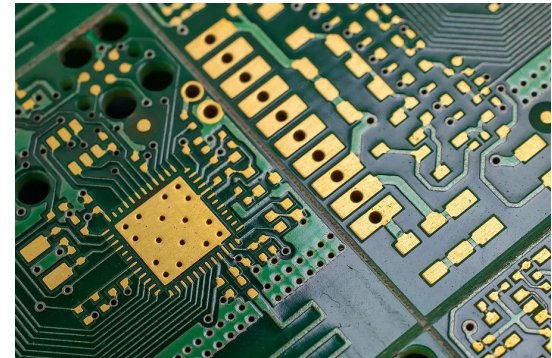
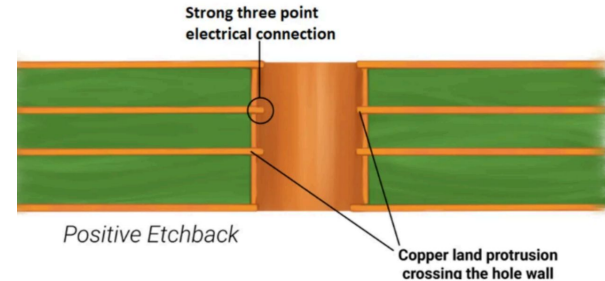
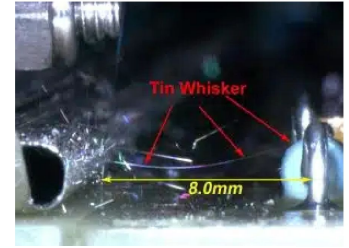
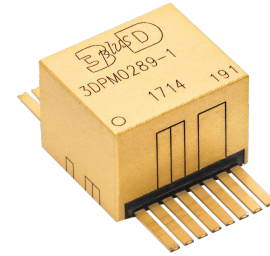


6 - Space Overview and application (SPENVIS)

- Simple radiation analysis of STM32H743VIT, COTS, 2 year mission:
 - 40 nm technology on SRAM → $0.2\ \mu\text{m} \times 0.2\ \mu\text{m} \times 0.2\ \mu\text{m}$ sensitive volume approx
 - 1 MB RAM contains 8388608 bits
 - SSO:
 - Long-term (SEU):
 - 2 mm of Al **910 errors**, 4 mm of Al **246 errors**
 - Short-term (SEU):
 - 2 mm of Al **1653 errors**, 4 mm of Al **451 errors**
 - MARS:
 - Long-term (SEU):
 - 2 mm of Al **2858 errors**, 4 mm of Al **761 errors**
 - Short-term (SEU):
 - 2 mm of Al **3235 errors**, 4 mm of Al **614 errors**
- SSO requires more Al protection while MARS requires more active mitigation and redundancy.

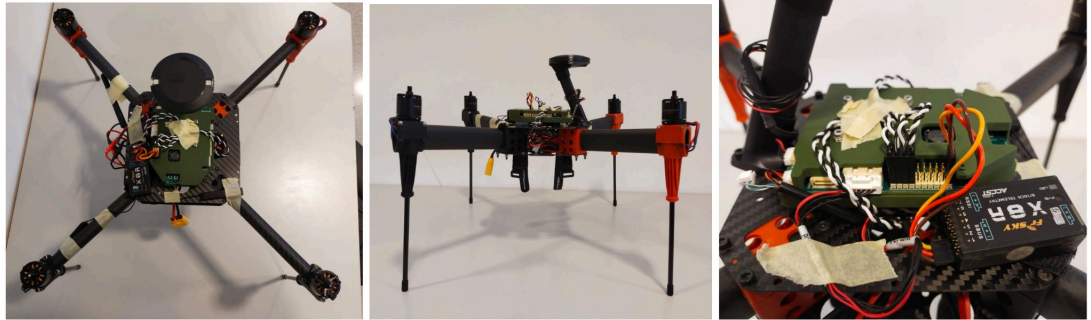
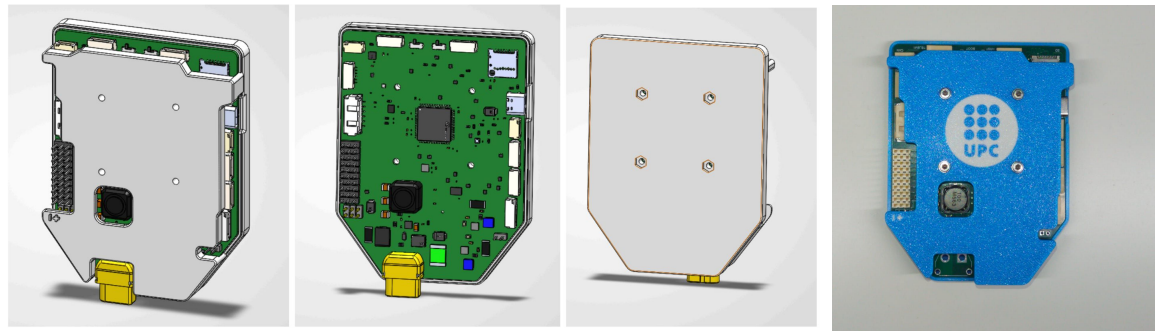
6 - Space Overview and application

- Multiple Standards analyzed:
 - **IPC, MIL-STD, NASA, ECSS.**
- Current COTS components considerations:
 - Derating and staking.
 - Bake-out campaign (outgassing).
 - Temperature, pressure, rad, vibration, shock tests.
 - LCL (Latch-up Current Limiters).
 - High Tg materials, with symmetric PCB.
 - SnPb (Tin whiskers) (solder and PGA/BGA/LGA).
 - ENIG (Electroless Nickel Immersion Gold).
 - ESD protection and positive etchback.
 - Special coating against arcing or corona effect.
 - Other: If possible Flip-chip package with PIND (Particle Impact Noise Detection) test.



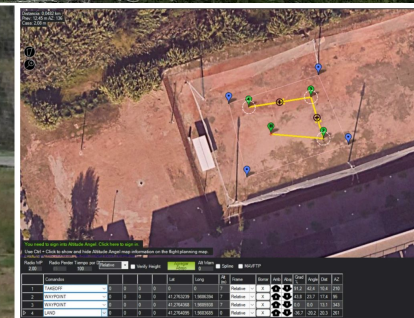
7 - Testing

- Simple printed case
- Hexsoon EDU 450
- Motor/ESCs kit
- Here 4 GNSS+Mag.
- Radio receiver.
- Sik telemetry radio.
- Some Mission Planner configuration.
- Additionally: due to ESD, HSE was broken (detected via RCC memory bit flip with STM32cube Programmer and voltage measure, 3 V instead of 1.65 V) → replacement.



7 - Testing

- 3 main tests:
 - Drone anchored.
 - Drone inside a geofence.
 - Autonomous mission.
- All tests were performed successfully.
- Remaining issues:
 - SDMMC inactive.
 - Check hwdef or crosstalk.
 - Noisy internal magnetometer.
 - PCB design → Impossible.
 - Advanced ESD protection and redundancy.



- Both magnetometer broken at internal level. Fixed Y or X value depending on the board, possible reason → Manufacturing or stress.

8 - Comparison

- Designed board:
 - Bridge between professional and hobbyist flight controller.
 - **Pros:**
 - High computational capability. Small and powerful STM32H7 (480 MHz).
 - Pixhawk connectivity. JST-GH.
 - Advanced electrical protection (ESD, reverse current, advance regulators, fuse, etc).
 - Precision timing architecture. MEMS/TCXO.
 - Advanced firmware integration. ArduPilot porting.
 - **Cons:**
 - Legacy sensors but more reliable.
 - No dedicated I/O co-processor.
 - Lack of multiple peripherals.
 - No PWM and ADC electrical protection. Lack of eFuse (LCL) on datalines.
 - Size and weight 77g.

9 - Cost and environment

- Cost:
 - Prototype: 186.77 € per board
 - 10 board production line: 65-70 € per board
- Environment and social impact:
 - All components are RoHS compliant.
 - Project addresses SDG (Sustainable Development Goals):
 - 9 - Industry, Innovation, and Infrastructure
 - 4 - Quality Education

Material	Consumption (approx)	Hours of consumption (approx)	Cost approx (0.15 €/kWh)	Co2 produced approx (0.246 kg/kWh)
Laptop	100 W	+600 h	9	14.76
		TOTAL:	9 €	14.76 kg

10 - Conclusions

- All objectives have been completed:
 1. Develop a functional drone control PCB.
 2. Create the drivers to manage the drone control PCB with ArduPilot.
 3. Evaluate the current design and expose space requirements.
- Project value:
 - Custom board porting:
 - Unlike Betaflight, ArduPilot required defining a low-level architecture.
 - Manually implemented the hwdef file to map STM32 resources to the ChibiOS.
 - Full open-engineering project:
 - Moving beyond the “Ardupilot Open Hardware” (limited to PDFs, hwdef, and in some quite rare cases, Gerber files).
 - Providing methodology: calculations, impedance, routing, layer stack-up, Altium files and porting steps.
 - Considerations:
 - Designed from scratch using reference architectures + space considerations.
 - Adapted circuits based on COTS stock availability and reduced cost.
 - Validated the theoretical design in a functional prototype, all in 5 months.

Thanks

Improvements

- Mandatory or mid-complex:
 - + Layers, change sensors, replace Mag to a Baro, coplanar lines, more redundancy and protection eFuse, double mounted side and new case.
- Optional and interesting:
 - Larger MCU package.
 - Add FRAM.
 - Apply multiple modules.
 - Ublox module or nRF module.
- I'm keen on:
 - Apply FPGA Xilinx or I/O co-processor.
 - Apply MCU K210 or AI analog IC with Raspberry in parallel.
 - Video buses with camera.

